

MIMIC-CTRL-2026: A Realistic Clinical Trial Dataset to Benchmark LLM performance

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Abstract

Patient recruitment remains a major barrier in clinical trials, as identifying eligible participants often requires manual screening of complex health records, which contributes significantly to trial delays and increased costs. While the advent of Large Language Models (LLMs) offers a chance to improve recruitment efficiency, existing public datasets used for development and evaluation are overly simplistic, failing to capture real world complexity. To address this, we introduce a novel dataset, MIMIC-CTRL-2026, composed of 800 manually labeled patient-clinical trial criterion pairs using real patient data. We show that models effective on simple datasets fail to generalize to this more complex data, highlighting inherent limitations in naive LLM approaches. Lastly, we present a novel clinical trial matching framework leveraging an LLM to successfully structure complex, unstructured eligibility criteria. Our method achieves strong performance without resource-intensive model fine-tuning.

Our dataset and code and full prompts are available at: *redacted for review*.

1 Introduction

Clinical trials are essential to ensuring safe and effective healthcare innovations (Friedman et al., 2015). Patient recruitment remains one of the most resource-intensive and time-consuming steps in clinical trials (Coorevits et al., 2013; Cowie et al., 2017). Identifying eligible participants often requires manual screening of complex health records, a challenge that contributes significantly to trial delays and costs (Huang et al., 2018). This cost is not equitably distributed – data extraction is more complex and costly when screening older patients (Sedrak et al., 2021), those with intellectual and developmental disabilities (higher co-morbidity burden) (McDonald et al., 2022) and children (higher rare disease burden) (US Food and Drug Administration, 2019).

A 51-year-old woman is seen in clinic for advice on osteoporosis. She has past medical history of significant hypertension and diet-controlled diabetes mellitus. She currently smokes 1 pack of cigarettes per day. She was documented by previous LH and FSH levels to be in menopause within the last year. She is concerned about breaking her hip as she gets older and is seeking advice on osteoporosis prevention.

Figure 1: An example patient case (from the SIGIR dataset) which is used to match or exclude patients based on trials’ inclusion and exclusion criteria.

Recent advances in large language models (LLMs) offer transformative potential to streamline clinical trial workflows, including automated interpretation of eligibility criteria, natural language search across heterogeneous data sources, and adaptive support tools for trial teams. Previous studies have demonstrated that AI-assisted patient matching can improve the process of enrollment (Unlu et al., 2025).

Unfortunately, the sensitive nature of the data required to develop and evaluate these methods creates hurdles for reproducibility. Much of the past research is conducted on private data which makes direct comparisons practically impossible. To address this, the research community has generated some public datasets for patient trial matching.

Unfortunately, the currently publicly available datasets are overly simplistic in their representation of patients. For example, a shared task from SIGIR-2016 (Koopman and Zuccon, 2016) represents patients as a single paragraph as illustrated in Figure 1. This overly simplifies the task of patient matching as real-world textual medical records for an individual patients would span multiple documents and formats (e.g., admissions notes, care notes, discharge notes, lab values, etc).

The over-simplicity of these datasets results in an over-estimation in the performance of LLM-based methods for patient-trial matching. To address these limitations, the contributions of our work are as follows:

- We introduce MIMIC-CTRL-2026, a manually annotated dataset of 800 patient-clinical trial criterion pairs for patient-trial matching using real-world complex healthcare data. The annotations were obtained from clinically trained annotators (a resident physician and a medical student supervised by a clinician-clinical trialist).
- We benchmark models on existing simple (single-paragraph) vs. realistic patient data, quantifying the performance gap.
- We present an end-to-end framework for automated eligibility matching that combines structured parsing, targeted extraction, and LLM-based criterion evaluation.
- We perform qualitative evaluation of the errors observed and reasoning provided highlighting some of the complexities surrounding patient matching as well as core limitations of existing datasets and approaches.

2 Related Work

The subsections below discuss related works on the following topics: a) existing public patient-clinical trial matching datasets, and b) existing work using LLMs for patient-trial matching.

2.1 Public Patient-Clinical Trial Matching Datasets

NLP research for patient-trial matching uses a variety of public and private datasets. While private datasets offer high ecological validity, they hinder reproducibility because they contain personally identifiable information (PII) which prevent them from being shared with the general research community. Thus many studies leverage public datasets which we discuss below:

2.1.1 ‘JV Health Search’ (SIGIR)

The first dataset, ‘JV Health Search’ (Koopman and Zuccon, 2016) (henceforth: SIGIR) is a dataset developed for the express purpose of evaluating information retrieval models’ performance at the task of patient matching. In this set-up patients

serve as the query to be matched with relevant trials. Patients (n=60) are represented using a description (average of 78 words) and a short summary (an average of 22 words) adopted from previous work (Simpson et al., 2014). Trials are represented using the inclusion and exclusion criteria from real-world clinical trials found at *clinicaltrials.gov*. A total of approximately 4000 manually annotated patient-trial pairs are available from this dataset.

2.1.2 Text REtrieval Conference (TREC) (2021–2023)

The next three datasets were all published as part of the Clinical Trials Track of the Text REtrieval Conference (TREC) in 2021, 2022, and 2023.

The first two of these datasets (henceforth TREC-2021 and TREC-2022) use the same underlying corpus (Roberts et al., 2022). With the task framed again as information retrieval, patients are topics and are represented using a short paragraph similar to what would be observed in an admission note. A few patient profiles have some selected information formatted as a list (often of lab values) appended to the prose. The 2021 iteration of the task had 75 patient profiles which was reduced to 50 in the 2022 iteration. Like the SIGIR dataset, inclusion and exclusion criteria from *clinicaltrials.gov* were used. Annotators ranked patient-trial pairings (n=35,832) using one of three labels: Not Relevant, Excluded, or Eligible.

TREC-2023, while very similar to the previous two iterations, instead of patients being represented using synthetic patient cases written in full prose, patients are instead represented using a simulated questionnaire that would match what the patient of their physician would fill out to identify eligible trials (TREC Biomedical Tracks Organizers, 2023).

2.1.3 n2c2

Unlike the previously described datasets, the n2c2 2018 shared task: *Cohort selection for clinical trials* (Stubbs et al., 2019) uses real-patient data. Each patient (n=288) is represented by 2-5 notes (with an average of 2711 tokens/patient). Each patient was manually annotated to see if they met a pre-selected set of 13 criteria extracted from real-studies also from ClinicalTrials.gov with minor modifications to maximize the utility of the patients.

2.1.4 Limitations of Public Datasets

While these works have made substantial contributions to the field, they share several limitations

that restrict their applicability to real-world clinical trial-patient matching. First, the SIGIR, TREC 2021, and 2022 datasets rely on synthetic paragraphs, substantially oversimplify the task and fail to reflect the complexity of real clinical narratives. Although the TREC 2023 track represents a realistic patient checklist, requiring the filling of such a checklist does not address the main bottleneck faced by clinicians which needs us to deal with the natural, longitudinal clinical text. Second, the n2c2 dataset, while more realistic in that it is derived from non-synthetic clinical data, remains limited in scope, as it: a) includes only a small number of discharge summaries per patient (typically two to five), and b) excludes other clinically relevant modalities, such as imaging reports, additional note types, lab tables, and medications. More importantly, the n2c2 dataset is no longer publicly available, further limiting its utility for reproducible and extensible research.

2.2 Automated Approaches to Trial Matching

Researchers have framed the task of trial matching in different ways. Researchers from the information retrieval sub-field frame the task as an information retrieval task where the patient is a query and is meant to be matched with all eligible trials (Roberts et al., 2022; Simpson et al., 2014; TREC Biomedical Tracks Organizers, 2023; Rybinski et al., 2024). In such framing, models are evaluated using metrics such as average precision@10, recall@10 and other retrieval metrics (Truong et al., 2022; Lahiri et al., 2024).

Researchers in NLP often frame the task as a classification problem where, given a patient’s profile and the inclusion and exclusion criteria for a specific trial, the model would classify whether the patient is eligible, ineligible, or not relevant to the particular trial. Such approaches tend to report traditional classification metrics (e.g., F1 score) (Tai and Tannier, 2025).

Some researchers combine both retrieval and classification in a multi-stage application (where retrieval is conducted to gather relevant trials followed by matching and sometimes a ranking as well) (Nievas et al., 2024; Jin et al., 2024).

For our work, we will focus on studies which have a classification component and highlight some particular relevant research. Jin et al. (2024), propose *TrialGPT*, a three-stage zero-shot framework built on OpenAI’s GPT-4 and GPT-3.5 models to match patients to clinical trials. The system first

generates trial-relevant keywords to enable large-scale retrieval, efficiently filtering tens of thousands of active trials (e.g., over 23,000 U.S. trials). In the second stage, the model evaluates patient eligibility at the individual inclusion and exclusion criterion level, producing natural-language explanations for its decisions. Finally, these criterion-level predictions are aggregated into trial-level eligibility scores. TrialGPT achieves near-expert performance, reporting an accuracy of 87.3% while reducing screening time by 42.6%.

Despite these promising results, TrialGPT relies primarily on synthetic and highly simplified patient data, which does not adequately reflect the complexity of real-world data which contain dense, heterogeneous, and temporally fragmented information, posing substantial challenges for direct LLM processing, retrieval, and matching.

Wornow et al. (2024) present a zero-shot LLM-based clinical trial screening system that achieves state-of-the-art performance without any task-specific training data. Their approach combines strategic prompting with a two-stage retrieval pipeline that first filters relevant clinical notes before applying eligibility assessment. The system reports an 81% Macro-F1 score and reduces screening costs by approximately 95%, addressing key practical challenges related to data scarcity, scalability, and deployment feasibility. However, this approach also has notable limitations. It is evaluated primarily on the n2c2 benchmark (Mahajan et al., 2023), which has multiple limitations: it focuses only on inclusion criteria, and is substantially smaller and less complex than real-world electronic health record (EHR) systems.

Chowdhury et al. (Chowdhury et al., 2024) proposed Siamese-PTM, a Siamese Neural Network using Llama 2 embeddings to model the semantic correlation between patient EHR and trial criteria. This architecture achieved a significant 40% performance gain over a rule-based baseline. Guluzade et al. (Guluzade et al., 2025) (ELMTEx) implemented fine-tuning on smaller open-source LLMs to optimize the extraction of structured clinical entities (diagnoses, treatments, etc.) from reports. Their primary limitation is that their system is designed for general clinical entity extraction using a custom dataset, lacking validation on a complex, established benchmark like MIMIC-IV and not addressing the intricate, holistic reasoning required for clinical trial eligibility assessment.

Motivated by the strengths (e.g., zero-shot ca-

pability, explainability) and limitations (e.g., reliance on synthetic data or restricted benchmarks) of these studies, we aimed to curate a publicly available dataset for patient-trial matching that would address these limitations. In doing so, upon discovery of poor generalization of existing methods to real-world data, we also developed a framework that operates robustly on non-synthetic, real-world patient data. Our methodology addresses the shortcomings of prior work by tackling the significantly more complex task of criterion-level patient-trial matching on the raw, noisy, and lengthy MIMIC-IV dataset. Specifically, we leverage Llama-70b-instruct to perform traceable, explainable reasoning across multiple patient documents. Rather than relying on simplified benchmarks, we evaluate multiple LLMs under different experimental configurations to systematically analyze their performance, robustness, and suitability for realistic clinical trial matching scenarios.

3 Creating MIMIC-CTRL

Current datasets oversimplify the patient trial matching process. In contrast, real-world clinical data is inherently messy, noisy, and voluminous, with a single patient profile comprising extensive narrative notes, structured tables, and medication histories. To address this gap, we introduce MIMIC-CTRL, a dataset featuring real-world patient records annotated against three trials. This dataset includes a total of 70 patient-trial pairs annotated per criterion ($n = 34$ inclusion and exclusion criteria), for a total of 800 expert-annotated patient-criterion pairs.

The deidentified patient records for this study were sourced from the MIMIC-IV dataset version 3.1 and the MIMIC-IV-Note module version 2.2¹². A cohort of 5,525 unique heart failure patients was identified by filtering for International Classification of Diseases (ICD) codes, specifically ICD-10 (starting with I50) and ICD-9 (starting with 428). For these patients, free-text discharge summaries and radiology reports were extracted alongside structured longitudinal data, including prescriptions and laboratory results for BNP and NT-proBNP. Because structured echocardiography data were unavailable during the extraction phase, these measurements were sourced directly from the

Inclusion Criteria: *Established chronic heart failure (≥ 6 months)*

Exclusion Criteria: *Unable to undergo CMR imaging or brain MRI*

Figure 2: Sample inclusion and exclusion criteria.

clinical notes. Additionally, medication data were refined to focus specifically on heart failure management, capturing relevant prescriptions for 5,270 patients within the cohort.

To optimize the manual annotation process, the extracted data were consolidated into a single unified source per patient. Discharge and radiology notes were concatenated in chronological order, placing the most recent information first, without truncation to maintain the integrity of the medical narrative. Structured vitals from online medical records were summarized to include the median, mean, and most recent values for systolic and diastolic blood pressure, as well as the most recent BMI. This organized format integrates concatenated notes, medication durations, blood pressure summaries, and relevant lab values into a comprehensive patient file. The resulting dataset was designed to facilitate the efficient review of complex eligibility criteria by providing a high-fidelity view of each patient’s clinical history. This streamlined approach minimizes data fragmentation by ensuring all relevant clinical indicators are presented in one cohesive document, and is illustrated in Appendix Figure 4.

3.1 Clinical Trials

In clinical trials, inclusion and exclusion criteria are conditions which must be met (or not met) by patients to qualify them for trials. Sample inclusion and exclusion criteria are presented in Figure 2.

Given the medical expertise of the annotators who partook in this task, we limited our analysis to three cardiology trials: CONVERGE-HF³ which investigates vericiguat in heart-failure patients with concomitant mild-to-moderate cognitive impairment, FINALITY-HF which investigates the efficacy and safety of finerenone in select patients with heart failure intolerant to or ineligible for steroidal mineralocorticoid receptor antagonists therapy⁴ and COMET(Poole-Wilson et al., 2002)

¹<https://physionet.org/content/mimiciv/3.1/>

²<https://physionet.org/content/mimic-iv-note/2.2/>

³www.clinicaltrials.gov/study/NCT06601465

⁴www.clinicaltrials.gov/study/NCT06033950

which investigates whether beta-blocking agents with differing pharmacological profiles exert different effects on morbidity and mortality in patients with chronic heart failure. The inclusion and exclusion criteria for all of these trials can be found in Appendix Tables: 8, 9 and 10.

3.2 Annotation Procedure

For the three trials we randomly sampled 30, 20, and 20 patients for annotation for Converge-HF, COMET, and FINALITY respectively. Manual annotation was performed by: a resident physician and a medical student under the supervision of a clinician-clinical trialist.

For the annotation schema, we had 4 labels, illustrated in Figure 4 – 0: “*The patient fails to meet the criteria OR has a condition that disqualifies them*”, 1: “*The patient meets the required criteria OR does not have the forbidden condition*”, 2: “*The medical records do not contain enough data to determine if the patient qualifies*”, 3: “*The specific criterion is irrelevant to this patient’s clinical context*”.

While the LLM could be provided all the data, our annotators found it unwieldy to read through the notes. Thus, to enable manual annotation of patient records we performed the following steps. For discharge notes, all notes were combined into a single document in the order of their occurrence (most recent being at the top). Radiology reports for the same patient cohort were extracted and processed in a manner analogous to the discharge notes. Prescription records were filtered to include only patients who were prescribed a wide set of drugs related to heart failure and the inclusion exclusion topics as defined by the clinical experts. Vital signs (e.g., blood pressure, patient weight, etc) were extracted and summaries were presented per visit (e.g., minimum, maximum, average, median) for both systolic and diastolic pressures. Specific lab values (related to the inclusion and exclusion criteria) were extracted on the request of the annotators (e.g., ejection fraction).

For our annotation methodology, first the medical student labeled all patient-criterion pairs. This was then reviewed by the resident physician who highlighted disagreements. Consensus was reached over the disagreements. Where needed, the supervising trialist was consulted to adjudicate between the two annotators.

4 Pre-Processing Datasets

Each of the datasets had to undergo pre-processing to normalize the data into a shared format to enable experimentation. At a high level, patient-level data was extracted and matched with the relevant trial data. More details can be found in Appendix Section B.

5 Patient Matching Experiments

We hypothesized that simple prompting, while effective on simple data, would fail on more realistic data. Below, we describe the experiments performed. Experiments use Llama-3.3-70B-Instruct unless stated otherwise. Prompts are shared in the Appendix.

5.1 Naive Prompt

Naive prompting is where the trial criteria is passed in with minimal modification as the prompt to an LLM with the patient information as context provided to the LLM. This evaluates model robustness when evaluated without trial-specific prompt customization or retrieval.

For SIGIR_2016, to match previous works, we construct a trial-agnostic prompt incorporating the trial ID, title, and criterion details. The model outputs JSON with a strict four-point eligibility score: “Inclusion” (meets inclusion or does not meet exclusion), “Not Included” (does not meet inclusion or meets exclusion), “Not Applicable” (e.g., pregnancy tests for males), or “Not Enough Information” (insufficient data). More technical details can be found in the appendix.

For TREC_2021, we applied the same set-up. We aggregated all inclusion and exclusion criteria for a specific patient–trial pair into a single comprehensive prompt to facilitate comparison with past work. Following past work, the model assigns one of three labels: non-relevant (0), excluded (1), or eligible (2). More details are in the appendix.

For MIMIC-CTRL, eligibility is categorized using a numerical scale: **1** denotes inclusion or non-exclusion, **0** denotes non-inclusion or exclusion, **2** indicates insufficient information, and **3** signifies that the criterion is not applicable. Prior to prompting, each clinical note was segmented into individual sentences and assigned unique sentence identifiers, which were included in the model input to enable evidence attribution in the generated outputs. The naive prompt for MIMIC-CTRL, presented in the Appendix, closely followed the structure used

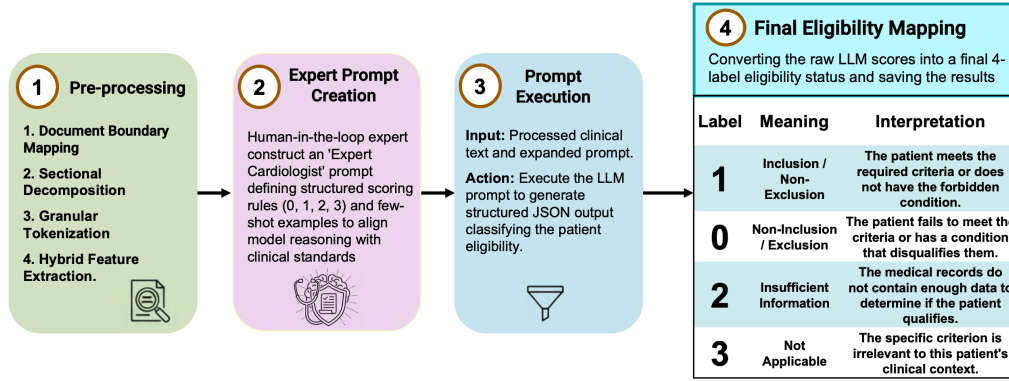


Figure 3: Illustration of Expanded Prompt-Processed Data methodology as applied to MIMIC-CTRL.

in SIGIR and TREC experiments, although the eligibility scoring scheme differed and was specifically designed by our annotators.

5.2 Naive Prompt w/ Processed Data (NP-PD)

For the MIMIC-CTRL dataset, we address the high token density of longitudinal patient records by implementing a hybrid, section-aware preprocessing pipeline. Rather than providing the model with raw, unstructured narratives, our system performs clinical segmentation to decompose discharge notes into standardized categories such as History of Present Illness and Discharge Medications.

The system utilizes a dual-pathway extraction strategy: deterministic regular expressions isolate individual summaries and extract key clinical indicators, such as ejection fraction and laboratory results, from headline-content pairs. For complex or unstructured entities that bypass regex patterns, we employ the LLM as a probabilistic fallback to ensure comprehensive data extraction. These filtered components are then concatenated into a refined representation, significantly reducing noise compared to the raw narratives. Finally, the processed data is indexed in a structured format, providing the LLM with a feature-rich input for accurate eligibility classification.

5.3 Expanded Prompt (EP)

Guided by clinical annotators, we developed a knowledge-enriched prompt, referred to as the *Expanded Prompt*. Without implementing external retrieval modules or fine-tuning the model, all clinical knowledge are specified directly as natural-language instructions in the system prompt. The prompt instructs the model to look into specific sections of the discharge summary, including *Brief*

Hospital Course, *History of Present Illness*, *Past Medical History*, and *Discharge Medications*, etc., when evaluating each eligibility criterion. For certain conditions, the prompt directs the model to search for lexical markers and clinical terminology, such as mentions of mild cognitive impairment or heart failure subtypes, within these sections. This section-aware instruction constrains the model’s reasoning to relevant portions of the note rather than searching indiscriminately across the entire document. Prompts were validated and iterated upon using two manually labeled patients (who were not part of the final MIMIC-CTRL-2026). Sample prompts are presented in the Appendix.

To capture clinical nuance, we specify a rule-based, multi-level scoring framework within the prompt. The framework defines four outcome labels: eligible, ineligible, insufficient information, and not applicable. Criterion-specific decision rules map observed evidence in the clinical note to these labels, including distinctions between prohibited and permissible but similar-sounding medications (e.g., *isosorbide dinitrate* versus *isosorbide mononitrate*). This prompt-specified structure enables accurate patient eligibility determination despite the absence of specialized pre-training on the clinical dataset.

5.4 Expanded Prompt w/Processed Data (EP-PD)

In our final experiment, we integrated the knowledge-enriched prompt architecture with the advanced data processing pipeline described in the preceding sections, Figure 3. This configuration yielded significantly higher success rates and improved the consistency of eligibility scoring across the dataset. The specific quantitative outcomes and

performance metrics associated with this optimized setup are detailed in the Results section.

Dataset	ACC	M-Prec	M-Rec	M-F1	Kappa
SIGIR	0.90	0.87	0.85	0.85	0.84
TREC	0.74	0.75	0.64	0.65	0.54
MIMIC*	0.28	0.33	0.34	0.25	0.06

Table 1: Performance of Naive Prompting on Llama model for MIMIC-CTRL. M-Prec: Macro Precision, M-Rec: Macro Recall, M-F1: Macro-F1, Kappa: Cohen’s Kappa.

5.5 Results

5.6 Quantitative Results

Evaluation Metrics: The performance of each model is evaluated using a suite of metrics designed to capture the nuances of multi-class classification. Accuracy provides a high-level measure of correctness, while Cohen’s Kappa measures the level of agreement between the model and the gold standard (taking random chance into account). To address the inherent class imbalance within clinical trial criteria, see confusion matrices in Appendix Figure 5, we present the macro-averaged precision, recall, and F1-scores to ensure that the model’s proficiency is weighed equally across all four categories, including minority classes such as ‘Not Enough Information’ and ‘Not Applicable’.

Cross-Dataset Performance Comparison: The experimental results in Table 1 demonstrate a significant performance degradation as the evaluation moves from simplistic/synthetic to real-world clinical data. While the model achieves near-perfect reliability on the SIGIR_2016 dataset with a Cohen’s Kappa of 0.84, this metric drops substantially to 0.54 for TREC_2021 (due to a more complex trial description) and collapses to 0.02 for the MIMIC_CTRL cohort. The sharp decline in Macro F1-scores from 0.85 to 0.25 highlights the model’s inability to both classify more nuanced labels, specifically for the complex logic required to identify ‘Not Enough Information’ and ‘Not Applicable’ cases and process noisy real-world data. Ultimately, these findings suggest that while naive prompting is effective for structured patient-trial data, it fails to provide clinically reliable eligibility decisions when faced with the high linguistic variance of real-world patient records.

Analysis of Model Performance: Table 3 reports the mean performance of the Llama-3.3-70B-Instruct model for all three trials, Appendix G

presents per-trial results. As expected, the NP model performed worst across all experiments. This is especially the case for exclusion criteria with a Kappa of 0.047 (barely better than random chance). Results indicate that processing the dataset using our described approach (indicated by _PD) is an effective approach, nearly doubling the performance for NP and EP (except for EP in exclusion criteria). The expanded prompt without specialized data processing (i.e., EP) – a form of expert-led prompt engineering – resulted in modest performance gains.

Model	Acc	M-Prec	M-Rec	M-F1	Kappa
Llama3.3-70B	0.50	0.50	0.42	0.42	0.26
Llama3.1-3B	0.46	0.39	0.35	0.34	0.12
Qwen72B	0.44	0.50	0.48	0.44	0.17
Qwen2.5-8B	0.33	0.44	0.37	0.29	0.11

Table 2: Performance of Expanded Prompting (EP-PD) for Llama on MIMIC-CTRL using various LLMs. M-Prec: Macro Precision, M-Rec: Macro Recall, M-F1: Macro-F1, Kappa: Cohen’s Kappa

Cross-Model Performance Evaluation:

The experimental results in Table 2, reveal a significant performance gap between architectures, with Llama3.3-70B emerging as the most capable model for processing these clinical records. In contrast, the Qwen72B model and the lightweight variants (3B and 8B) demonstrate limited effectiveness when utilizing the EP-PD framework, yielding considerably lower agreement scores. This suggests a “reasoning threshold” where a model must possess both high parameter scale and superior instruction-adherence to navigate the linguistic noise and intricate logic found in MIMIC-IV notes. The poor Macro F1-scores for smaller models specifically highlight an inability to distinguish definitive eligibility from nuanced categories like “not enough information.” While Qwen72B possesses a large parameter count, its failure to outperform the 70B Llama variant warrants further exploration.

5.6.1 Qualitative Results

The disagreements between annotators and the model were the result of numerous differences. The three most significant ones include: a) differences in judgment, b) differences in knowledge, and c) differences in the ability to identify key information in complex free-text. In the process of screening patients the models struggled at times to identify devices, conditions or medications. For example when evaluating a patient’s discharge sum-

Model	Inclusion Criteria					Exclusion Criteria				
	Acc.	M-Prec.	M-Rec.	M-F1	Kappa	Acc.	M-Prec.	M-Rec.	M-F1	Kappa
NP	0.377	0.417	0.336	0.314	0.109	0.228	0.305	0.368	0.218	0.047
NP_PD	0.634	0.516	0.484	0.461	0.456	0.371	0.399	0.374	0.319	0.106
EP	0.540	0.474	0.445	0.427	0.300	0.558	0.441	0.434	0.388	0.240
EP-PD	0.721	0.635	0.545	0.544	0.535	0.416	0.414	0.346	0.323	0.143

Table 3: Mean LLama3.3-70B-Instruct Performance on MIMIC-CTRL split by criteria. *Acc.*: (Overall Accuracy), *M-Prec.*: Macro Precision, *M-Rec.*: Macro Recall, *M-F1*: Macro F1-Score, *Kappa*: Cohen’s Kappa

mary the model identified sublingual nitroglycerin as a long acting nitrate, but this particular formulation is short acting and would not fulfill the criteria for exclusion in this patient’s example. In other instances the model correctly identified relevant factors considered by the annotator, but came to the incorrect conclusion. For example, it correctly identified that a patient had multiple cardiovascular comorbidities such as coronary artery disease and hypercholesterolemia, but incorrectly assumed that this suggested on its own that the patient may have fulfilled the criteria for having chronic heart failure which is a related, but separate condition.

Finally, the model showed some positive and some negative behaviours when searching for key information. There were numerous instances where the model effectively identified information which was missed by annotators in their review that ultimately led to the more correct determination by the model. There are also some instances of hallucinations by the model regarding the information contained in the patient documents, leading it to make incorrect conclusions.

The model’s explanations were usually correct when it had correctly classified the pairing. Explanations for incorrect predictions were somewhat incoherent and failed to address the core qualities of the criteria that determined if it would be fulfilled or not.

6 Discussion

Screening patients for participation in clinical trials is a challenging, time-intensive task. There is a clear role for model assisted screening in reducing the burden of trial screening particularly with complex well demarcated criteria that require little clinical interpretation. Interpretation of patient information and trial criteria is challenging even for clinicians; even with a consensus screening design determinations are marred with ambiguity. More specifically, during human annotations, disagreements in the screening process arose both from the identification of disparate information and conflict-

ing interpretations of identical data, underscored by the inherent complexity of clinical documentation. These variations are exacerbated by the ambiguity of eligibility criteria, which often lack standardized thresholds, such as specific hemoglobin levels for “severe anemia,” thereby allowing subjective clinical experience to dictate divergent interpretations.

The amount of information available to inform clinicians in trial screening decision-making is inversely proportional to the amount of ambiguity within these decisions. However, this volume of data presents specific challenges for automated systems; our results indicate that as the number of trial conditions increases, the Llama model’s performance begins to decline. Similarly, performance decreases as the overall length of the prompt increases, suggesting a sensitivity to information density and context window saturation. Even in unified electronic medical record systems, the completeness of documentation is variable and clinicians may be forced to follow up with patients and other care providers. For this reason, we disagree with [Stubbs et al. \(2019\)](#), which concluded that all instances of “possibly met” could be replaced with a binary decision. We believe that an intermediate determination such as “possibly” is the result of a cautious approach to uncertainty which maintains the integrity of clinical trials and reduces the risk of subjecting patients to preventable harms. In a non-clinical research setting where information availability has substantial limitations, such an intermediate determination is even more critical.

7 Conclusion

In this work, we introduced **MIMIC-CTRL-2026**, a realistic dataset for patient-trial matching. Our experiments revealed that LLM performance on synthetic data is over-estimated and doesn’t generalize to the complexity of real-world clinical records. Our EP-PD model setup highlights the gains from both prompt engineering as well as structured preprocessing, paving a new path for research on automated clinical trial matching.

8 Limitations

Because of the sensitive nature of the data, there are not many publicly available datasets which could be annotated to create our dataset. This has various impacts on our work. First, our specific findings of model performance are less robust to performance on other datasets – though we think the substantial difference in performance indicates that trends will hold but the degree may differ.

MIMIC is a dataset of patients admitted to the ICU which are more severe patients unlikely to be sought for the average cardiology trial (when compared to traditional cardiology outpatients). MIMIC also redacted some social information to minimize the risk of re-identification – information which could change a patients eligibility (and our model’s performance). Likewise, performance patterns (e.g., Llama outperforming Qwen) may change when applied to other datasets.

Given the clinical expertise of our collaborators, we limited ourselves to cardiology trials. Thus, our observed results may not hold to other medical specialties (e.g., trials from other specialties may have harder or easier criteria). However we believe that the core contributions of our work: highlighting that overly simplistic datasets result in over-estimation of model performance and the creation of the dataset are not affected by this limitation.

Non-trialists may question not using expert trialists to label data and the effect of such a decision on data quality. In the US most research coordinators do not have as much experience as medical students and residents. Thus, our decision is in-line with current standards for clinical trials run in the US.

Our work is also restricted to the English language (due to data availability), and the findings may not generalize well to other languages or multilingual clinical contexts. Our annotation of the dataset was manual, requiring expert physician annotators to go over entire patient files (which could span years) which limited the number of patient-trial pairs and the overall number of trials that could be annotated.

The application of LLMs to enable trial matching is generally low-risk. However, the technology to find people based on description’s from large, complex, and unstructured texts could be used in surveillance and other negative scenarios. While we don’t believe that our work directly contributes to this (i.e., the data and code won’t be useful for

surveillance purposes), learning gained (e.g., the framework developed, and general findings) could be used for nefarious actions. Unfortunately, we don’t think there is a way of preventing this while enabling our work. There are also other externalities to our work. For example, while less severe, this work required substantial time from physicians which theoretically could be spent providing care. Likewise, processing large amounts of data using LLMs may have negative effects on the environment.

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A Pre-processing Details

SIGIR_2016 From the publicly available SIGIR dataset⁵, we extracted the patient’s summary text from <TOP> and <NUM> tags, while trial metadata and eligibility criteria (both inclusion & exclusion) were parsed from the clinical_trial records. No heavy pre-processing was required, as each patient note consists of approximately 125 words, where we simply numbered each sentence. These separate sources were combined into a table format consisting of patient information and trial criteria paired row-wise. Ultimately, 1,016 unique patient-trial combinations were selected for the experimental evaluation.

TREC_2021 TREC provides 75 patient descriptions (referred to as “topics”⁶) and consists of approximately 31,000 expert-annotated patient-trial labels. From these, we selected a subset of 2,074 unique combinations due to computational limitations and the trial descriptions are more complex than that of SIGIR_2016. The patient notes and trial information were combined like the SIGIR dataset into a single table formatted specifically for the LLM eligibility experiments.

MIMIC The MIMIC-IV dataset contains detailed clinical narratives, including history of present illness, physical exams, and discharge medications, family history etc. Since patients often have multiple admissions, a single patient’s records can exceed 20–25,000 tokens. Our initial processing consistent of simply passing these raw, extensive notes directly into the LLM.

All datasets, codes, and models utilized in this study were collected and used according to their respective licenses and access requirements.

B Prompt, Evaluation and Technical Details

C Technical Details

C.1 SIGIR_2016

Inference uses greedy decoding with a temperature of 0.0, a 16,384-token context window, and a 512-token limit for new generations.

C.2 TREC_2021

We employed greedy decoding (temperature 0.0) within a 16,384-token context window to ensure deterministic reasoning and classification results.

⁵<https://data.csiro.au/collection/csiro:17152>

⁶<https://www.trec-cds.org/2021.html>

C.3 MIMIC

The prompt was limited to a maximum of 20,000 tokens (with irrelevant content trimmed if necessary), and inference was performed with a temperature of 0.3 and max_new_tokens set to 528. This configuration assessed the model’s robustness when evaluated against the CONVERGE-HF and COMET trial criteria using full clinical narratives, without trial-specific prompt customization or retrieval. Overall, this naive prompting strategy yielded poor performance, achieving classification accuracies of only 47.7% for inclusion criteria and 19.4% for exclusion criteria for the CONVERGE-HF trial (Table 8).

D Llama Model Details (General & Project-Specific)

The Llama-3.3-70B-Instruct model, a 70-billion-parameter large language model (LLM) developed by Meta, is typically deployed using bfloat16 or float16 precision with higher-temperature sampling to support creative text generation. In this project, however, the model was configured for efficient, resource-constrained multi-GPU inference by applying 4-bit NF4 quantization via the BitsAndBytes library in conjunction with float16 precision. This quantization strategy reduced the model’s VRAM requirements from approximately 140 GB to roughly 35–40 GB, enabling deployment on limited hardware resources. Furthermore, text generation was performed using a low temperature ($T = 0.05$) and predominantly greedy decoding (do_sample = False) to ensure consistent and deterministic outputs, which are critical for maintaining high fidelity in clinical reasoning tasks involving long and complex patient records.

E Compute

The experiments described in the paper were performed on NVIDIA A100-SXM4-40GB GPUs. All fine-tuning and inference experiments were conducted on 2-3 GPUs. Running all models took approximately 15-20 hours in total.

F Use of AI Assistants

An AI assistant was used only for spelling, grammar, and phrasing.

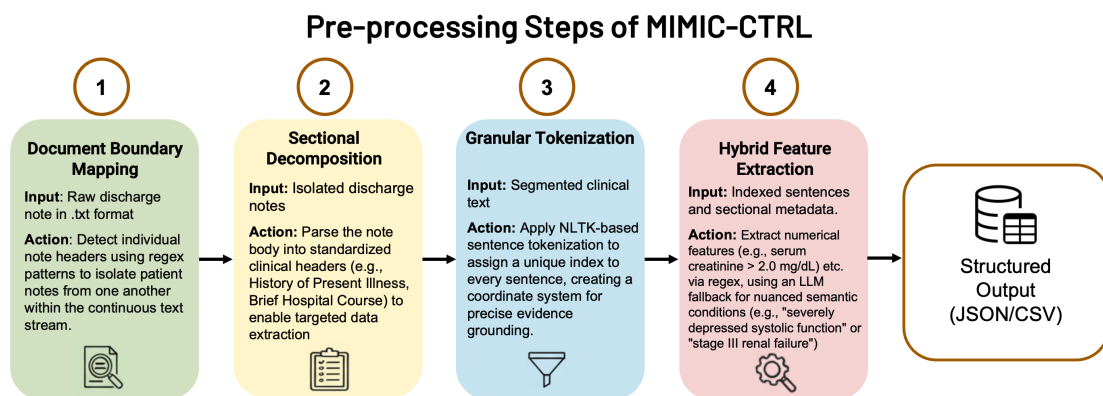


Figure 4: Illustration of Pre-Processing methodology for creating the MIMIC-CTRL dataset.

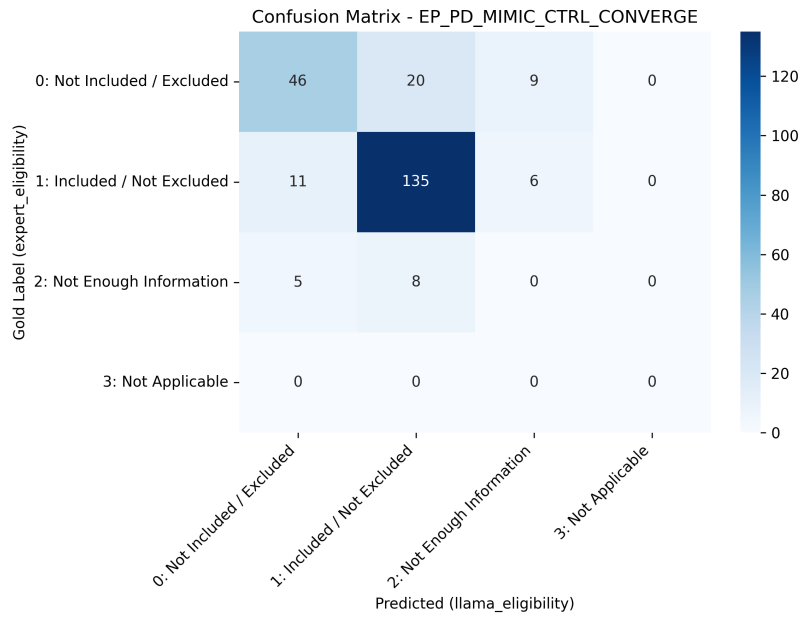
Table 4: Naive_Prompt_for_MIMIC_IV

Component	Content										
System Instruction	You are an expert physician matching patients to clinical trials. Evaluate the patient note against the following inclusion or exclusion criteria.										
Instructions	1. Determine whether the criterion is applicable to the patient. 2. Identify whether there is direct evidence supporting or contradicting the criterion. 3. If information is unclear or missing, label as <i>not enough information</i>.										
Output Format	Return only a JSON object with the keys: score, reasoning, and sentence_ids. The value of score must be one of 0, 1, 2, or 3.										
Patient Note	<i>[Token-limited patient clinical note formatted as sentence_id: sentence]</i>										
Criterion	<i>[Clinical trial inclusion or exclusion criterion]</i>										
Eligibility Scoring Rubric	<table> <tr> <th>Score</th><th>Interpretation</th></tr> <tr> <td>1</td><td>Meets inclusion or does not meet exclusion</td></tr> <tr> <td>0</td><td>Does not meet inclusion or meets exclusion</td></tr> <tr> <td>2</td><td>Insufficient information available</td></tr> <tr> <td>3</td><td>Criterion not applicable</td></tr> </table>	Score	Interpretation	1	Meets inclusion or does not meet exclusion	0	Does not meet inclusion or meets exclusion	2	Insufficient information available	3	Criterion not applicable
Score	Interpretation										
1	Meets inclusion or does not meet exclusion										
0	Does not meet inclusion or meets exclusion										
2	Insufficient information available										
3	Criterion not applicable										

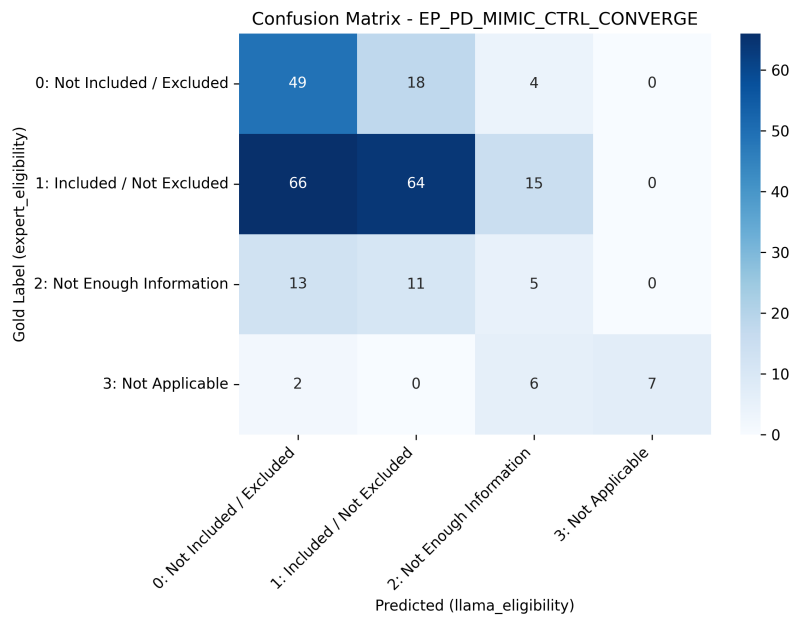
G Per-Trial Performance

In this section, we present the per-trial results. Table 11 presents the results for the CONVERGE-HF trial. Table 12 presents the results for the COMET Trial. Table 13 presents the results for the FINALITY trial.

H Confusion Matrix



(a) Confusion Matrix for the Llama3.3-70B-Instruct model using EP_PD on the CONVERGE-HF Trial.



(b) Confusion Matrix for the Llama3.3-70B-Instruct model using EP_PD on the COMET Trial.

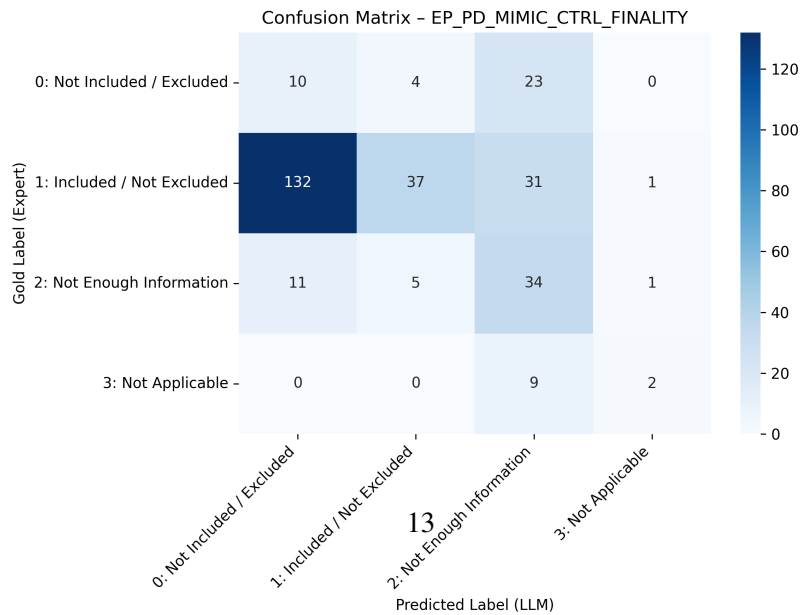


Table 5: Expanded Prompt for CONVERGE_HF Trial

Component	Content
System Instruction	You are an expert physician evaluating patient eligibility for a clinical trial. Assess the discharge note and determine whether the patient meets the specified inclusion or exclusion criterion, following trial-specific rules.
Operational Guidelines	1. Determine if the criterion applies to the patient. 2. Identify direct evidence supporting or contradicting the criterion. 3. If information is unclear or missing, label as <i>not enough information</i>. 4. Use keyword searches (case-insensitive, regex) in relevant sections: History of Present Illness, Past Medical History, Discharge Diagnosis, Medications, Brief Hospital Course, Past Cardiac Procedures. 5. Evaluate all conditions in the criterion without assuming inapplicability. 6. All patients are adults. Score “1” for the "Adult patients" criterion. 7. For time-sensitive criteria (e.g., “within 3 months”), the condition must appear in the most recent note under Discharge Diagnosis or History of Present Illness. 8. For criteria unlikely to be mentioned (e.g., allergies), score 1 if no explicit mention unless ambiguous.
Specific Criteria	• Adult patients: Score 1. Reason: All patients are adults, no note evidence needed. • Established chronic heart failure (≥ 6 months): Check History of Present Illness, Past Medical History, Discharge Diagnosis for keywords: chronic heart failure, CHF, HFrEF. Score 1 if found, else 2 if no data. Assume ≥ 6 months unless explicitly stated otherwise. • Mild-to-moderate cognitive impairment: Check Brief Hospital Course, Past Medical History, Discharge Diagnosis for MCI, Alzheimer’s Dementia, AD, cognitive impairment. Score 1 if any found, else 2 if no data. • Contraindications for sGC stimulator / vericiguat: Check Discharge Medications, Brief Hospital Course, Discharge Diagnosis for riociguat, sildenafil, tadalafil, vardenafil, hemoglobin < 8 g/dL, pregnancy, breastfeeding. Score 1 if none found, 0 if any contraindication found (except Isosorbide Mononitrate), 2 if ambiguous, 3 for pregnancy/breastfeeding in males. • Unable to undergo CMR / Brain MRI: Check Past Cardiac Procedures, Brief Hospital Course, Discharge Diagnosis for pacemaker, implantable device, Adapta. Score 0 if any found, 1 if none, 2 if no data. • CMR exclusions: Check Past Cardiac Procedures, Brief Hospital Course, Discharge Diagnosis for renal failure, CKD, chronic kidney disease, GFR < 30, ICD, PPM, CRT, Adapta. Score 0 if any found, 1 if none, 2 if no data. • General medical conditions: Check Brief Hospital Course, Past Medical History, Discharge Diagnosis for Alzheimer’s Dementia, AD, dementia, Basal Cell Cancer, cancer, malignancy. Score 0 if any found, 1 if none, 2 if no data. • Allergies to study products: Check History of Present Illness, Past Medical History for allergy, allergic. Score 0 if found, 1 if none, 2 if ambiguous.
Output Format	Return only a JSON object with keys: score, reasoning, sentence_ids. Example: <code>{{"score": "2", "reasoning": "...", "sentence_ids": []}}</code>

Table 6: Expanded Prompt for COMET Trial

Component	Content
System Instruction	You are an expert physician evaluating patient eligibility for a clinical trial. Assess the discharge note and determine whether the patient meets the specified inclusion or exclusion criterion, following trial-specific rules.
Operational Guidelines	Evaluate ONLY the given criterion. Follow the COMET guidelines strictly. Search medical terms case-insensitively. Reference sentence numbers where applicable. Return ONLY a valid JSON object with fields: "score", "reasoning", and "sentence_ids".
Specific Criteria	• NYHA Class II–IV: Look in: Active Issues, Brief Hospital Course. Search: "NYHA", "class II", "class III", "class IV", "dyspnea", "HFpEF", "syncope", "volume overload". Scoring: 1 (symptoms II), 0 (class I or none), 2 (insufficient). Reasoning: Matched terms + severity. • 1 CV Admission Past 2 Years: Look in: Brief Hospital Course, Discharge Diagnosis. Search: "NSTEMI", "CAD", "CABG", "valve replacement", "heart failure", "stroke". Scoring: 1 (1 event), 0 (none), 2 (timing unclear). Reasoning: List events + timeframe. • Stable HF Treatment: ACE inhibitor 4 weeks AND diuretic 2 weeks (40 mg furosemide equiv.). Look in: Discharge Medications. Search: "ACE inhibitor", "enalapril", "captopril", "lisinopril", "furosemide", "Lasix", "diuretic". Scoring: 1 (both stable), 0 (missing/inadequate), 2 (unclear). Reasoning: List drugs/doses/stability. • Cardiac Function (EF 0.35 in 3 mo OR LVEDD >6.0 cm OR FS <20%): Look in: Brief Hospital Course, Echocardiogram Findings, Past Medical History. Search: "EF", "LVEF", "LVEDD", "fractional shortening", "HFrEF", "HFpEF". Scoring: 1 (meets any), 0 (>0.35 or HFpEF), 2 (no recent data). Reasoning: Cite values + timing. • Recent Change in HF Treatment: Look in: Discharge Medications, Brief Hospital Course. Search: "amiodarone", "diltiazem", "verapamil", "metoprolol", "carvedilol", "amlodipine", "inotrope". Scoring: 0 (recent excluded), 1 (stable), 2 (timing unclear). Note: Assume recent hospitalization satisfies timing unless contradicted. Reasoning: List drugs + relevance. • Unstable Angina/MI/CABG/Stroke 2 Months: Look in: History of Present Illness, Brief Hospital Course, Discharge Diagnosis. Search: "unstable angina", "NSTEMI", "STEMI", "MI", "CABG", "stroke", "infarct". Scoring: 0 (recent), 1 (none), 2 (timing unclear). • Uncontrolled Hypertension (SBP >170 or DBP >105): Look in: Vital Signs. Scoring: 0 (uncontrolled), 1 (controlled), 2 (missing). Reasoning: Report BP values. • Hemodynamically Significant Valvular Disease: Look in: History of Present Illness, Past Medical History. Search: "AS", "valvular", "bioprosthetic", "severe". Scoring: 0 (significant), 1 (stable/treated), 2 (severity unclear). • Sustained VT/VF (<2 Months, Untreated): Look in: Brief Hospital Course, Discharge Diagnosis. Search: "VT", "VF", "ventricular", "ICD". Scoring: 0 (present), 1 (absent/controlled), 2 (unclear). • Pregnancy or Inadequate Contraception (Females Only): Look in: History of Present Illness. Scoring: 3 (male), 0 (pregnant/no contraception), 1 (adequate), 2 (unknown). • Drug/Alcohol Misuse, Poor Compliance, Serious Systemic Disease: Look in: Social History, Past Medical History, Brief Hospital Course. Search: "alcohol", "ETOH", "drug abuse", "non-compliance", "malignancy", "cancer", "terminal". Scoring: 0 (present), 1 (absent), 2 (unclear). • Poor Compliance or Serious Systemic Disease Complicating Management: Look in: Social History, Past Medical History, Brief Hospital Course, Discharge Diagnosis. Search: "non-compliance", "non-compliant", "poor compliance", "malignancy", "metastatic", "terminal illness", "end-stage", "dementia", "advanced COPD", "severe liver disease", "renal failure on dialysis". Scoring: 0 (present), 1 (none mentioned), 2 (insufficient). • Contraindication to -Blockers: Look in: Past Medical History, Active Problems, Discharge Diagnosis. Search: "bradycardia", "AV block", "COPD", "PAD", "asthma", "sick sinus syndrome", "bifascicular block". Scoring: 0 (present), 1 (none), 2 (unclear). Reasoning: Quote phrases or state absence.
Output Format	Return only a JSON object with keys: score, reasoning, sentence_ids. Example: {"score": "1", "reasoning": "...", "sentence_ids": [1, 5]}

Table 7: Expanded Prompt for nsMRA Trial (e.g., FINERENONE or Similar)

Component	Content										
System Instruction	You are an expert physician evaluating patient eligibility for a clinical trial. Evaluate whether the patient satisfies the given criteria type using ONLY the provided discharge note information.										
Operational Guidelines	• 1: Meets inclusion OR not excluded • 0: Does not meet inclusion OR excluded • 2: Insufficient or unclear information • 3: Not applicable (e.g., pregnancy criteria for male patients)										
Specific Criteria	• INCLUSION: Symptomatic HFrEF per protocol: Look in: Brief Hospital Course, History of Present Illness, Past Medical History. Positive: Explicit "HFrEF", reduced EF, systolic HF PLUS symptoms (dyspnea on exertion, orthopnea, NYHA II–IV). Scoring: 1 (both present), 0 (HFrEF stated or missing component), 2 (unclear). • INCLUSION: Not on steroidal MRA due to intolerance/contraindication/ineligibility: Look for: spironolactone, eplerenone, MRA. Scoring: 1 (explicitly avoided due to intolerance/contraindication), 0 (currently on sMRA), 2 (no mention). • INCLUSION: Pregnancy-related criteria: Scoring: 3 (male patient); For females — 1 (documented testing/contraception), 2 (undocumented). • EXCLUSION: Treatment with non-steroidal MRA (nsMRA): nsMRA: finerenone, esaxerenone. Scoring: 0 (explicitly treated), 1 (explicitly not treated or no mention). • EXCLUSION: eGFR <25 mL/min/1.73m² OR potassium >5.0 mmol/L: Use explicit labs only. CKD Stage III implies eGFR >30 → 1. Scoring: 0 (threshold breached or CKD IV/ESRD mentioned), 1 (acceptable), 2 (unclear). • EXCLUSION: MI, revascularization, valve surgery, or CRT within 30 days or planned: Scoring: 0 (recent or planned), 1 (remote history only), 2 (dates unclear). • EXCLUSION: Prior or planned heart transplant: Scoring: 0 (documented), 1 (not planned or no mention). • EXCLUSION: Severe uncorrected valvular disease as primary HF cause: Requires cardiology attribution. Scoring: 0 (clearly primary cause), 1 (no indication), 2 (severity/causality unclear). • EXCLUSION: Symptomatic bradycardia or heart block WITHOUT pacemaker: Scoring: 1 (bradycardia/heart block with PPM present OR no info), 0 (bradycardia/heart block without pacemaker). • EXCLUSION: Excluded cardiomyopathies (inflammatory, infiltrative, HOCM, etc.): Scoring: 0 (explicitly present), 1 (no indication). • EXCLUSION: Probable alternative cause of HF: Scoring: 0 (explicitly stated), 1 (no indication). • EXCLUSION: Concomitant potent CYP3A4 inhibitors or moderate/potent inducers: Review discharge medications only. Scoring: 0 (present), 1 (none). • EXCLUSION: Any other condition making patient unsuitable (e.g., lung cancer, liver failure): Scoring: 0 (explicitly stated), 1 (no indication). • EXCLUSION: Concurrent participation in another interventional trial: Scoring: 0 (documented), 1 (no information).										
Output Format	Return only a JSON object with keys: score, reasoning, sentence_ids. Example: {"score": "1", "reasoning": "...", "sentence_ids": [1, 5]}										
Eligibility Scoring Rubric	<table> <thead> <tr> <th>Score</th><th>Interpretation</th></tr> </thead> <tbody> <tr> <td>1</td><td>Meets inclusion or does not meet exclusion</td></tr> <tr> <td>0</td><td>Does not meet inclusion or meets exclusion</td></tr> <tr> <td>2</td><td>Insufficient information</td></tr> <tr> <td>3</td><td>Not applicable (e.g., pregnancy in males)</td></tr> </tbody> </table>	Score	Interpretation	1	Meets inclusion or does not meet exclusion	0	Does not meet inclusion or meets exclusion	2	Insufficient information	3	Not applicable (e.g., pregnancy in males)
Score	Interpretation										
1	Meets inclusion or does not meet exclusion										
0	Does not meet inclusion or meets exclusion										
2	Insufficient information										
3	Not applicable (e.g., pregnancy in males)										

Table 8: Clinical Trial Criteria: CONVERGE-HF

Component	Content
Inclusion Criteria	• Adult patients? • Established chronic heart failure (≥ 6 months) • Mild-to-moderate cognitive impairment (as per the diagnosis of cognitive impairment or a Montreal Cognitive Assessment (MoCA) score 10-25).
Exclusion Criteria	• Patients who have contraindications for sGC stimulator and vericiguat therapy (i.e. use of long-acting nitrates, other soluble guanylate cyclase stimulators (e.g., riociguat), or phosphodiesterase type 5 (PDE-5), severe anemia, pregnancy or breast-feeding) • Unable to undergo CMR imaging or brain MRI. • CMR exclusions: renal failure [a glomerular filtration rate <30 mL/min)], implantable cardiac device (ICD or CRT), uncontrolled atrial fibrillation or recurrent ventricular arrhythmias). • General medical conditions: uncontrolled thyroid disorders, hepatic failure, or myocardial revascularization procedures [coronary angioplasty and/or surgical revascularization in the previous 3 months], cancer/malignancy, or with moderate-severe dementia). • Patients with allergies to the study products.

Table 9: Clinical Trial Criteria: FINALITY

Component	Content
Inclusion Criteria	• Age $\geq$ 18 years or legal age of majority • Symptomatic heart failure with reduced ejection fraction (HFrEF) per protocol-defined criteria • Not receiving steroidal mineralocorticoid receptor antagonists (sMRA) due to prior intolerance, contraindication, or ineligibility for treatment • Negative pregnancy test and agreement to use adequate contraception during the trial (female participants only)
Exclusion Criteria	• Treatment with non-steroidal mineralocorticoid receptor antagonists (nsMRA) • Estimated glomerular filtration rate (eGFR) < 25 mL/min/1.73 m² and/or serum or plasma potassium > 5.0 mmol/L • Type 1 acute myocardial infarction, coronary revascularization, valve replacement or repair, or implantation of a cardiac resynchronization therapy device within 30 days prior to enrollment or planned during the study • Prior or planned heart transplantation • Hemodynamically significant (severe) uncorrected primary cardiac valvular disease considered by the investigator to be the primary cause of heart failure • Symptomatic bradycardia or second- or third-degree heart block in the absence of a pacemaker • Cardiomyopathy due to known acute inflammatory heart disease, infiltrative diseases, accumulation diseases, muscular dystrophies, cardiomyopathy with reversible causes, hypertrophic obstructive cardiomyopathy, complex congenital heart disease, or pericardial constriction • Probable alternative cause of the participant's heart failure • Concomitant systemic therapy with potent cytochrome P450 isoenzyme 3A4 (CYP3A4) inhibitors or moderate to potent CYP3A4 inducers • Any other condition or therapy that would make the participant unsuitable for the study • Concurrent participation in another interventional clinical study using an investigational agent

Table 10: Clinical Trial Criteria: COMET

Component	Content
Inclusion Criteria	• New York Heart Association [NYHA] class II–IV • At least one cardiovascular admission during the previous 2 years • On stable heart failure treatment with ACE inhibitors for at least 4 weeks unless contraindicated, and on treatment with diuretics (≥ 40 mg of frusemide or equivalent) for at least 2 weeks. • Left-ventricular ejection fraction had to be 0.35 or lower measured within the previous 3 months by echocardiography or radionuclide ventriculography. If the ejection fraction was not determined, a left-ventricular end diastolic diameter of greater than 6–0 cm and a fractional shortening of less than 20% as measured by echocardiography could qualify patients..
Exclusion Criteria	• Major exclusion criteria were a recent change of treatment—defined as the introduction of a new class of drug for heart failure or treatment with oral β-adrenergic or α-adrenergic receptor blockers within the 2 weeks before randomisation—requirement for intravenous inotropic therapy, current treatment with calcium channel blockers (of the diltiazem or verapamil class), amiodarone (> 200 mg per day) or class-I antiarrhythmic drugs, or administration of any investigational drug (specifically metoprolol or carvedilol) within the preceding 30 days. • Excluded patients with unstable angina, myocardial infarction (STEMI, NSTEMI, MI), or coronary revascularisation (CABG) or stroke within the previous 2 months. • Other exclusion criteria were uncontrolled hypertension (blood pressure systolic >170 mm Hg or diastolic >105 mm Hg). • Haemodynamically significant valvular disease • Symptomatic and sustained ventricular arrhythmias within the past 2 months not adequately treated with antiarrhythmic drugs or implantation of an automatic defibrillator. • Pregnancy, women with childbearing potential on inadequate contraception. • Known drug or alcohol misuse. • Poor compliance with treatment or any other serious systemic disease that might complicate management and reduce life expectancy. • Excluded patients if there was a contraindication to use of β-blockers, such as resting heart rate < 60 beats per min, sick sinus syndrome, bifascicular block, second or third degree atrioventricular block unless treated with a pacemaker, sitting systolic blood pressure < 85 mm Hg, history of asthma or chronic obstructive pulmonary disease (COPD), peripheral arterial disease (PAD) with symptoms at rest, or unstable insulin-dependent diabetes mellitus.

Model	Inclusion Criteria					Exclusion Criteria				
	Acc.	M-Prec.	M-Rec.	M-F1	Kappa	Acc.	M-Prec.	M-Rec.	M-F1	Kappa
NP	0.46	0.51	0.30	0.37	0.19	0.21	0.28	0.34	0.18	-0.03
NP_PD	0.79	0.60	0.48	0.52	0.64	0.28	0.30	0.25	0.20	-0.02
EP	0.44	0.52	0.39	0.36	0.21	0.71	0.44	0.41	0.41	0.24
EP-PD	0.80	0.57	0.56	0.56	0.64	0.73	0.49	0.46	0.47	0.37

Table 11: Mean LLama3.3-70B-Instruct Performance on MIMIC-CTRL split by criteria from CONVERGE_HF. Acc: (Overall Accuracy), M-Prec: Macro Precision, M-Rec: Macro Recall, M-F1: Macro F1-Score, Kappa: Cohen’s Kappa

Model	Inclusion Criteria					Exclusion Criteria				
	Acc.	M-Prec.	M-Rec.	M-F1	Kappa	Acc.	M-Prec.	M-Rec.	M-F1	Kappa
NP	0.35	0.49	0.39	0.35	0.12	0.39	0.45	0.52	0.37	0.22
NP_PD	0.55	0.37	0.33	0.35	0.31	0.44	0.40	0.48	0.37	0.24
EP	0.69	0.54	0.55	0.54	0.47	0.69	0.54	0.55	0.54	0.47
EP-PD	0.68	0.57	0.56	0.56	0.46	0.39	0.51	0.39	0.40	0.10

Table 12: Mean LLama3.3-70B-Instruct Performance on MIMIC-CTRL split by criteria from COMET. *Acc:* (Overall Accuracy), *M-Prec:* Macro Precision, *M-Rec:* Macro Recall, *M-F1:* Macro F1-Score, *Kappa:* Cohen’s Kappa

Model	Inclusion Criteria					Exclusion Criteria				
	Acc.	M-Prec.	M-Rec.	M-F1	Kappa	Acc.	M-Prec.	M-Rec.	M-F1	Kappa
NP	0.32	0.25	0.22	0.23	0.01	0.09	0.18	0.24	0.1	-0.05
NP_PD	0.56	0.58	0.64	0.51	0.42	0.39	0.51	0.39	0.40	0.10
EP	0.49	0.36	0.40	0.38	0.22	0.27	0.35	0.34	0.21	0.01
EP-PD	0.69	0.77	0.51	0.51	0.51	0.13	0.25	0.20	0.11	-0.04

Table 13: Mean LLama3.3-70B-Instruct Performance on MIMIC-CTRL split by criteria from Finality. *Acc:* (Overall Accuracy), *M-Prec:* Macro Precision, *M-Rec:* Macro Recall, *M-F1:* Macro F1-Score, *Kappa:* Cohen’s Kappa